

2-8 Adjust Numbers to Add or Subtract

Learning goal: I can change the numbers in an addition or subtraction problem to friendlier numbers and keep the sum or difference the same.

Textbook: pages 63–66. The book adjusts $224 + 109$ to $223 + 110$ and $333 - 212$ to $331 - 210$.

Words to Know

adjust	Change the numbers a little to make them easier to work with.
friendly number	A number that is easy to add or subtract, like 200 or 110.
addend	A number you add.
sum / difference	The answer when you add / the answer when you subtract.
equation	A number sentence with an equal sign, like $223 + 110 = 333$.

How to Teach It

- Find the number that is close to a ten or hundred (like 198 or 109).
- Adding:** take some away from one addend and give the same amount to the other (**opposite** moves). $224 + 109 \rightarrow 223 + 110$.
- Subtracting:** add the same amount to both numbers, or take the same amount from both (**same** moves). $333 - 212 \rightarrow 331 - 210$.
- Show subtraction on a number line: sliding both numbers the same amount keeps the distance between them.
- If only one number was changed, fix the answer: $645 + 300 = 945$ used 2 too many, so take 2 away $\rightarrow 943$.

Tip: Use coins for addition (move one coin from one pile to the other: the total stays the same). Ask, “Adding or subtracting? Opposite or same?”

Worked Examples

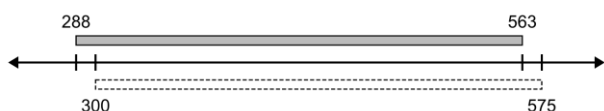
Example 1: Adjust addition (opposite moves)

$$\begin{array}{rclcl} 236 & + & 198 & = & ? \\ -2 & & +2 & & \\ \downarrow & & \downarrow & & \\ 234 & + & 200 & = & \mathbf{434} \end{array}$$

Example 2: Adjust subtraction (same moves)

$$\begin{array}{rclcl} 452 & - & 197 & = & ? \\ +3 & & +3 & & \\ \downarrow & & \downarrow & & \\ 455 & - & 200 & = & \mathbf{255} \end{array}$$

Example 3: Why subtraction uses the same moves



$563 - 288 \rightarrow 575 - 300 = \mathbf{275}$. Both moved up 12, so the distance stays the same.

Example 4: Mistakes to fix

$367 + 196 \rightarrow 370 + 199$? **No.** Both went up 3, so the sum is 6 too big. Use $363 + 200 = \mathbf{563}$.

$645 + 298$: changed only 298 to 300 $\rightarrow 945$. Added 2 extra, so take 2 away: **943**.

Skills Checklist *Check off each skill your child can do without help.*

- | | |
|---|--|
| ☐ Adjust an addition problem by moving an amount from one addend to the other. | ☐ Show on a number line that the difference stays the same. |
| ☐ Adjust a subtraction problem by changing both numbers the same way. | ☐ Tell if someone adjusted correctly. |
| ☐ Explain how adjusting addition and subtraction are different. | ☐ Fix the answer when only one number was changed. |

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Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** Adjust to make a friendly number. Write the change on each arrow. Then add.

$$\begin{array}{r} 248 \\ \hline \downarrow \end{array} + \begin{array}{r} 195 \\ \hline \downarrow \end{array} = ?$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- B.** Adjust to make a friendly number. Write the change on each arrow. Then subtract.

$$\begin{array}{r} 364 \\ \hline \downarrow \end{array} - \begin{array}{r} 198 \\ \hline \downarrow \end{array} = ?$$

$$\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- C.** Rosa adds $538 + 207$. She adjusts it to $540 + 209$ and finds 749. Do you agree with her strategy? Explain.

On My Own *Try these by yourself. Show your work.*

- 1.** Adjust the equation to solve.

$$\begin{array}{r} 463 \\ \hline \downarrow \end{array} - \begin{array}{r} 98 \\ \hline \downarrow \end{array} = ?$$

$$\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- 2.** Adjust the equation to solve.

$$\begin{array}{r} 156 \\ \hline \downarrow \end{array} + \begin{array}{r} 237 \\ \hline \downarrow \end{array} = ?$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- 3.** Adjust the equation to solve.

$$\begin{array}{r} 769 \\ \hline \downarrow \end{array} - \begin{array}{r} 412 \\ \hline \downarrow \end{array} = ?$$

$$\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- 4.** Adjust the equation to solve.

$$\begin{array}{r} 499 \\ \hline \downarrow \end{array} + \begin{array}{r} 286 \\ \hline \downarrow \end{array} = ?$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

5. Adjust the equation to solve.

$$672 - 318 = ?$$

↓

↓

$$\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

6. Adjust the equation to solve.

$$148 + 159 = ?$$

↓

↓

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

7. Kim changed $208 + 119 = ?$ to $207 + 120 = ?$. Describe how she adjusted the equation. Why do you think she did it that way?

8. Show how you can adjust $584 - 297$. Use the number line to show that the new equation has the same difference.



New equation: _____

9. You can adjust $345 - 128$ in different ways. Explain one way and why it makes the problem easier.

10. Tia and Sam find $139 + 268$ by adjusting. Tia uses $140 + 269 = 409$. Sam uses $140 + 267 = 407$. Which sum is correct? Explain.

11. A library had 674 books last year. Now it has 935 books. How many more books does it have now? Adjust the numbers to solve.

More books: _____

12. Jon adds $624 + 199$. He changes 199 to 200 but forgets to change 624. He gets 824. How can he fix his sum? Explain.

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Together

A	Sample: -5 and $+5 \rightarrow 243 + 200 = \mathbf{443}$ <i>Accept any adjustment that takes from one addend and gives the same amount to the other (e.g., $250 + 193 = 443$).</i>
B	Sample: $+2$ and $+2 \rightarrow 366 - 200 = \mathbf{166}$ <i>Both numbers must change the same way (e.g., $364 - 198 \rightarrow 366 - 200$).</i>
C	No. She added 2 to both numbers, so her sum is 4 too big. In adding, one number goes up and the other goes down: $545 + 200 = \mathbf{745}$.

On My Own

1	$+2, +2 \rightarrow 465 - 100 = \mathbf{365}$ <i>Items 1–6: accept any correct adjustment. Addition: opposite moves. Subtraction: same moves.</i>
2	$-3, +3 \rightarrow 153 + 240 = \mathbf{393}$
3	$-2, -2 \rightarrow 767 - 410 = \mathbf{357}$
4	$+1, -1 \rightarrow 500 + 285 = \mathbf{785}$
5	$+2, +2 \rightarrow 674 - 320 = \mathbf{354}$
6	$-1, +1 \rightarrow 147 + 160 = \mathbf{307}$
7	She took 1 from 208 and gave it to 119. Now 120 is a friendly number, and the sum stays the same: $207 + 120 = \mathbf{327}$.
8	Add 3 to both: $587 - 300 = \mathbf{287}$. On the number line, both ends slide 3 to the right, so the distance stays 287. <i>Accept any correct same-move adjustment drawn on the line (e.g., $584 - 297 \rightarrow 587 - 300$).</i>
9	Sample: add 2 to both $\rightarrow 347 - 130 = \mathbf{217}$. It is easier because 130 is a friendly number. <i>Accept any same-move adjustment, e.g., $342 - 125 = 217$ or $350 - 133 = 217$.</i>
10	Sam (407). Tia added 1 to both numbers, so her sum is 2 too big. Sam took 1 from 268 and gave it to 139.
11	$935 - 674$. Sample: take 4 from both $\rightarrow 931 - 670 = \mathbf{261}$ more books. <i>Also accept $961 - 700 = 261$ (add 26 to both) or any correct method.</i>
12	He added 1 too many (200 instead of 199), so he should subtract 1: $824 - 1 = \mathbf{823}$.